

4. Rising gas prices are affecting many households. The government can help households keep their gas bills as low as possible.

Data for a typical house are shown below. The table also shows government estimates for the current usage of each method.

Part of house	Heat loss before insulation is fitted (%)	Method of reducing heat loss	Cost of method (£)	Annual saving on gas bills (£)	% of homes already using this method
roof	25	loft insulation	450	150	91
windows & doors	10	double glazing	3 900	90	70
walls	40	cavity wall insulation	1 200	250	42
floor	15	carpets	800	40	90
draughts	10	draught excluders	60	5	85

- (a) A heating engineer estimates keeping the house with no insulation at 20°C results in gas bills of £2 000 per year.

Use the information above to answer the following questions.

- (i) Calculate the cost of the heat lost **through the roof**. [2]

cost = £

- (ii) State how the engineer's estimate will change if the home is kept at 22°C. [1]

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- (iii) State how the engineer's estimate will change if the price of gas drops. [1]

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- (iv) The engineer recommends that loft insulation should be installed first.

Calculate the payback time for loft insulation. [2]

payback time = years

- (v) Which method of reducing heat loss should the government help people install by giving them a grant? Give **one** reason for your choice. [2]

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- (b) At present, many homes with central heating lack a timer, sufficient hot water tank insulation, room thermostat and thermostatic radiator valves. Upgrading all homes so they have these four features could make savings of over 25% of the energy used for heating and hot water.

Fitting new hot water tank insulation, a room thermostat and thermostatic radiator valves reduces the annual gas bill for a house from £2 000 to £1 400.

- (i) Calculate the percentage saving made to the gas bill. [2]

% saving =

- (ii) Use the equation:

$$\text{decrease in the number of units used} = \frac{\text{savings}}{\text{cost of one unit}}$$

to calculate the decrease in the number of units used. [2]
Assume each unit costs £0.20.

decrease in the number of units used = kWh

- (c) State **two** ways in which using less gas to heat the house will benefit the environment. [2]

1.

2.